



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2024

MATHEMATICS (N510)

TIME: 2 HOURS

Candidates answer on the Question Paper.
Additional Materials: Geometrical Instruments

READ THESE INSTRUCTIONS FIRST

- 1. Write your index number in the space provided above.
- 2. Write in dark blue or black ink.
- 3. You may use an HB pencil for diagrams. Do not use correction fluid.
- 4. Diagrams are not drawn to scale unless otherwise specified.
- 5. Answer **ALL** questions.
- 6. All workings should be shown in the spaces provided.
- 7. This document consists of **33** questions printed on **19** pages, numbered **2** to **20**.
- 8. Calculators must **NOT** be used for this paper.
- 9. The number of marks is given in brackets [] at the end of each question or part question.
- 10. The total number of marks for this paper is **100**.

For Examiners' use												
Page No.	3	5	7	9	11	13	15	17	19	20	Total	Signature
Examiner												
Team Leader												
CE/ACE												

1. Work out:

$$\begin{array}{r} 632 \\ + 157 \\ \hline \end{array}$$

Answer: [1]

2. Evaluate:

$$\frac{5}{13} + \frac{3}{13}$$

Answer: [1]

3. Convert 5 kg 620 g into grams.

Answer: g [1]

4. Find $\sqrt[3]{64}$.

Answer: [1]

5. Simplify $(m^3)^5$.

Answer: [1]

6. Circle the **square** number in the list below.

15

20

36

59

80

[1]

7. Simplify $7x^2 + 2x^2$.

Answer: [1]

8. Evaluate $3 + 5 - 9$.

Answer: [1]

9. Express 0.31 as a percentage.

Answer: % [1]

10. Calculate 3.02×4

Answer: [1]

11. Factorise $3x - 6$.

Answer: [1]

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12. Circle the correct answer. Each item carries 1 mark.

(a) How many right angles are there in a rectangle?

A 2

B 3

C 4

D 5

(b) What is the value of 2 in 502.6?

A 2 hundreds

B 2 tens

C 2 tenths

D 2 units

(c) A clock shows 6.00 p.m. The clock is 20 minutes **ahead of time**. What is the correct time?

A 5.20 p.m.

B 5.40 p.m.

C 6.20 p.m.

D 6.40 p.m.

(d) 130 minutes =

A 1 h 30 min

B 1 h 40 min

C 2 h 10 min

D 2 h 30 min

(e) What is the highest common factor (**H.C.F.**) of $3x$ and $6x^2$?

A $3x$

B $6x$

C $3x^2$

D $6x^2$

(f) Which of the following is a **prime number**?

A 21

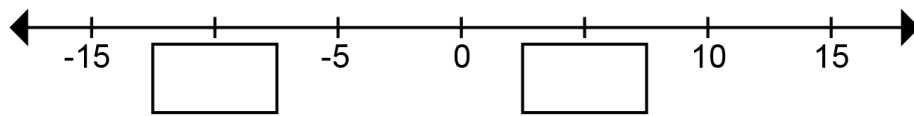
B 23

C 25

D 27

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13. Complete the number line below.



[1]

14. (a) Arrange the following numbers in **ascending** order.

5.62 5.06 5.23 5.65 5.006

Answer:,,,, [1]

- (b) After a discount of 10%, a laptop costs Rs 27 900.
Find the **original price** of the laptop.

Answer: Rs [2]

15. Solve the following equations:

(a) $x - 5 = -7$

Answer: $x = \dots\dots\dots$ [1]

(b) $x^2 - 5x + 6 = 0$

Answer: $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [2]

16. (a) Solve the inequality $2x + 3 \leq 11$.

Answer: $\dots\dots\dots$ [2]

(b) Draw a number line to represent the inequality in **part (a)**.

[1]

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17. In a class of 30 students,
14 play football,
17 play volleyball,
 x play both football and volleyball,
6 play neither football nor volleyball.
- (a) By using a Venn diagram or otherwise, find the value of x .

Answer: $x = \dots\dots\dots$ [3]

- (b) How many students play volleyball only?

Answer: $\dots\dots\dots$ students [2]

18. Using the information given below, find $\sqrt{5000}$.
 $[\sqrt{5} = 2.236, \sqrt{50} = 7.071]$

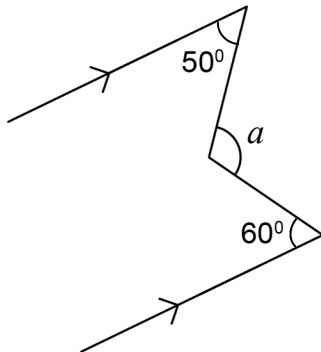
Answer: [2]

19. Insert brackets into this statement to make it true:

$$50 \div 2 \times 5 - 4 = 1$$

[1]

20. Find angle a .



Answer: angle $a =$ ^o [2]

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21. (a) Kylie runs a distance of $1\frac{3}{4}$ km. She then cycles a distance of 3500 m. Finally, she swims a distance of 2250 cm. What is the **total distance** covered by Kylie, in **metres**?

Answer: m [3]

- (b) A man invests a sum of Rs 300 000 for 5 years at the rate of 2% per annum. Calculate the **simple interest** the man earns.

Answer: Rs [2]

22. (a) Make ' a ' the subject of formula.

$$3b = \frac{2a - 1}{4 - 5a}$$

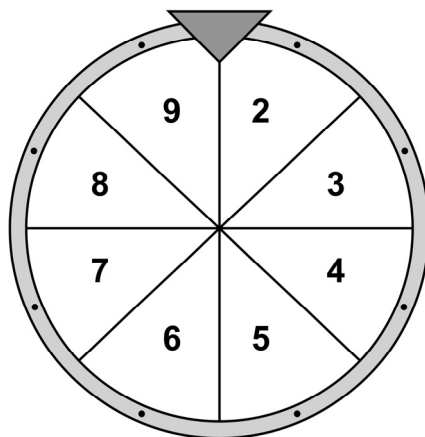
Answer: [3]

- (b) If $A = 6m + \frac{7n}{100}$, find the value of A when $m = 2$ and $n = 3$.

Answer: $A =$ [2]

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23. The spinner shown below, numbered 2 to 9, is spun **once**.



- (a) Find the probability that the arrow shows an **even number**.

Answer: [1]

- (b) Find the probability that the arrow **does not** show '7'.

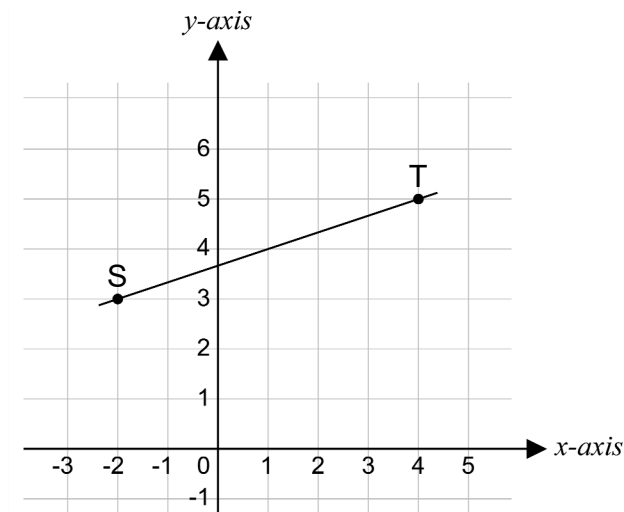
Answer: [1]

24. Given that $A = \begin{pmatrix} 2 & -3 \\ 0 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 4 \\ 6 & 2 \end{pmatrix}$,

Find $2A - B$.

Answer: [2]

25. (a) The graph below shows the line passing through S (-2 , 3) and T (4 , 5).



Find the equation of the line passing through (-3 , 1) and parallel to the line ST.

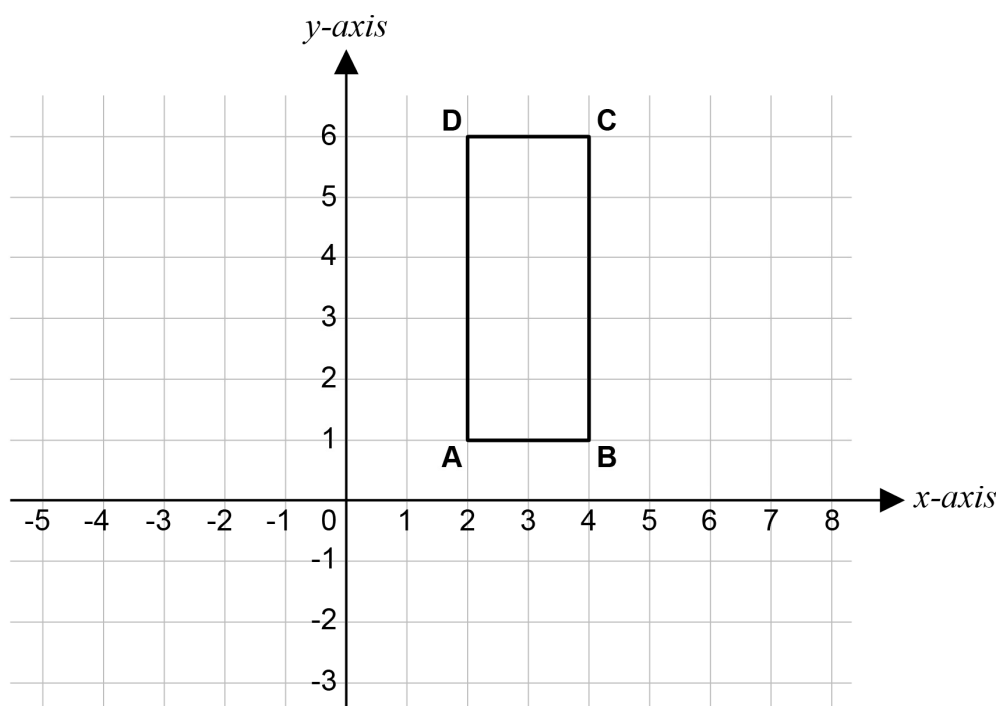
Answer: [4]

- (b) 12 men take 15 days to build a wall.
How many **more** men are needed to build the same wall in 10 days?
[Assume that all men work at the same rate]

Answer: men [3]

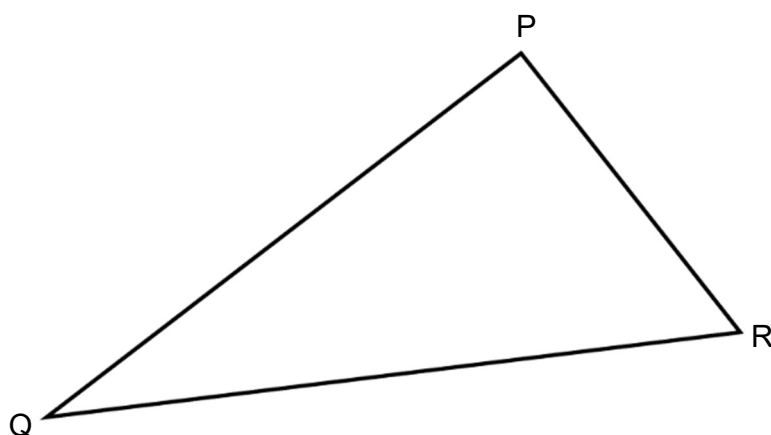
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26. (a) Rectangle **ABCD** has vertices **A** (2 , 1), **B** (4 , 1), **C** (4 , 6) and **D** (2 , 6).



- (i) Translate rectangle **ABCD** onto **A₁B₁C₁D₁** by $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$. [2]
- (ii) Reflect rectangle **ABCD** onto **A₂B₂C₂D₂** in the *y*-axis. [2]

- (b) Consider triangle PQR.



Using a pair of compasses,

- (i) construct the perpendicular bisector of QR. [1]
- (ii) construct the angle bisector of angle QPR. [1]

27. Matchsticks are arranged to form a pattern as shown below.

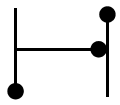


Figure 1

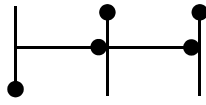


Figure 2

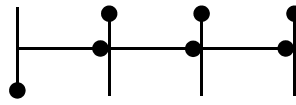


Figure 3

.....

Figure 4

(a) Draw **Figure 4** to continue the pattern.

[1]

(b) The table below shows the number of matchsticks used in each figure.

Figure	1	2	3	...	5	...	b	...	n
Number of matchsticks	3	5	7	...	a	...	21	...	c

(i) Find the value of a .

Answer: $a = \dots\dots\dots$ matchsticks [1]

(ii) Find the value of b .

Answer: $b = \dots\dots\dots$ [1]

(iii) Write c in terms of n .

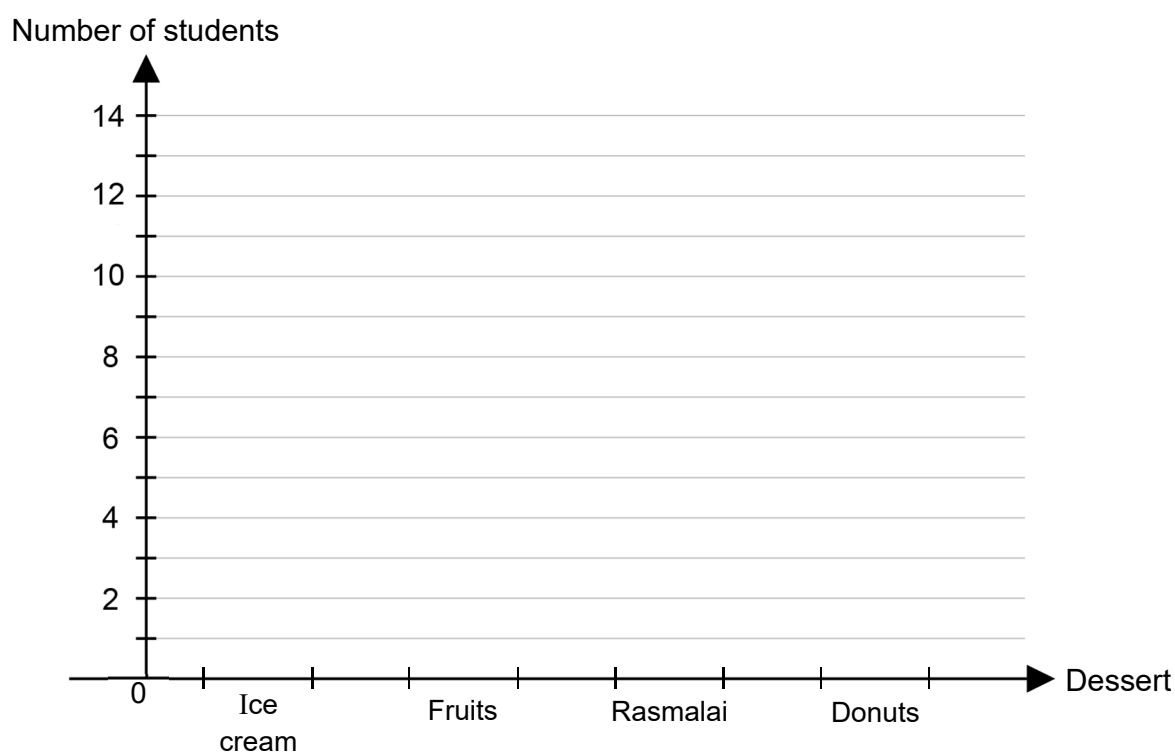
Answer: $\dots\dots\dots$ [1]

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28. A teacher conducts a survey to find out the favourite dessert of 36 students. The results are shown in the table below.

Dessert	Number of students
Ice cream	13
Fruits	10
Rasmalai	7
Donuts	6

- (a) Draw a bar chart to represent the above information.



[2]

- (b) If the information given in the table is represented on a pie chart, calculate the angle that represents the number of pupils whose favourite dessert is **Fruits**.

Answer:^o [2]

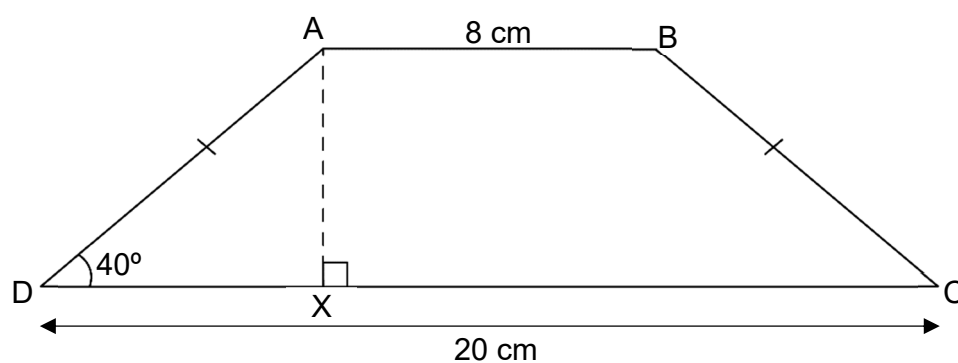
29. ABCD is a trapezium, where $AD = BC$.

Angle $AXC = 90^\circ$

Angle $ADX = 40^\circ$

$AB = 8 \text{ cm}$ and

$CD = 20 \text{ cm}$.



Calculate length AX .

$[\cos 40^\circ = 0.77, \sin 40^\circ = 0.64, \tan 40^\circ = 0.84]$

Answer: cm [4]

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30. A sum of money is shared between Shania and Mirella in the ratio 2 : 5.
Mirella receives Rs 360 **more than** Shania.

Calculate the **total amount of money** shared.

Answer: Rs [3]

31. A pair of shoes and 3 T-shirts cost Rs 3750.
Two pairs of the same shoes and 5 T-shirts cost Rs 6750.

Find the cost of 1 pair of shoes and 4 T-shirts.

Answer: Rs [6]

32. A circle has radius 13 cm.

- (a) Find the circumference of the circle. **[Leave your answer in terms of π]**

Answer: cm [2]

- (b) A triangle is found inside the circle as shown.

$OA = 13$ cm
 $AB = 10$ cm and
 $\angle OXA = 90^\circ$

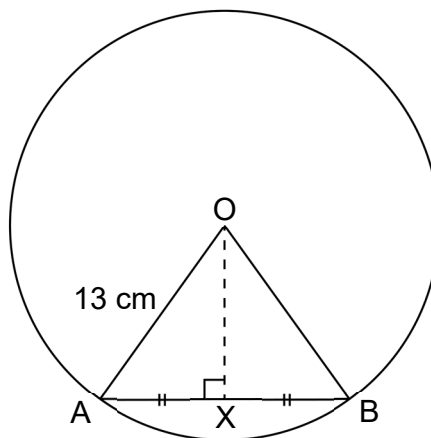


Diagram not to scale

- (i) What type of triangle is triangle AOB?

Answer: [1]

- (ii) Calculate the length of OX.

Answer: cm [3]

- (iii) Calculate the area of triangle OAB.

Answer: cm^2 [2]

33. The length of each side of a square is increased by 8 cm.
The area of the **new** square is 6 times its **perimeter**.

Find the length of one side of the **original square**.

Answer: cm [7]

END OF PAPER

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